National University of Singapore

CS1101S — Programming Methodology

AY2022/2023 Semester 1

Final Assessment

Time allowed: 2 hours

ANSWER SHEET

STUDENT NUMBER

A

(1) [3 marks]

```
function insert_last(C, x) {












}
```

(2) [4 marks]

```
function make_RCL(L) {
    let C = pair(head(L), null); // new pair for the first element.





    return C;
}
```

(3) [4 marks]

```
function append_RCLs(C1, C2) {




}
```

(4) [5 marks]

```
function remove_last(C) {
    // Find second last pair.
    let second_last = C;
    while (tail(second_last) !== C) {

    }

    return second_last;
}
```

(5) [4 marks]

```
function RCL_to_stream(C) {
   function helper(D) {
      return
   }
   return helper(tail(C));
}
```

(6) [3 marks]

```
function find_2nd_min(bmt) {
  return
}
```

(7) [5 marks]

```
function find_max(bmt) {

  return is_empty_tree(left_branch(bmt))
         && is_empty_tree(right_branch(bmt))
         ?
}
```

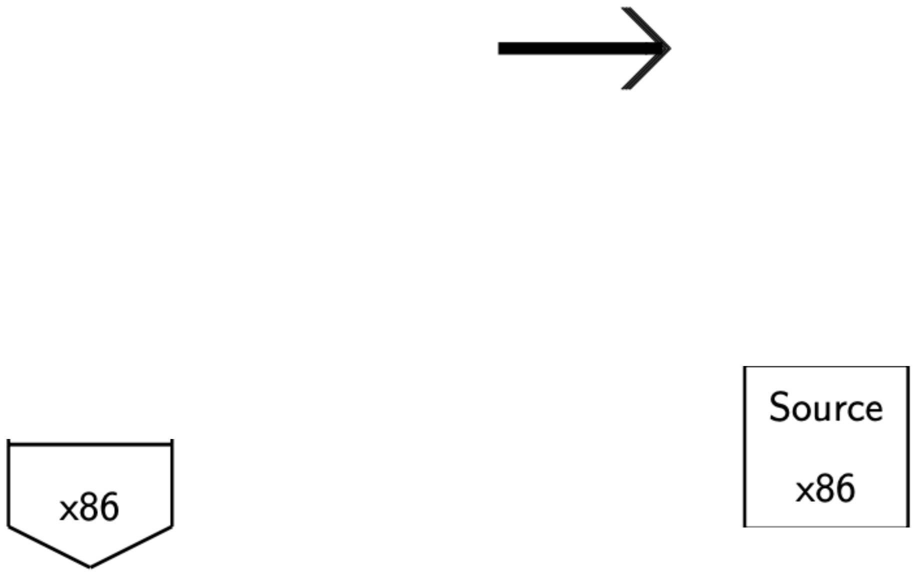
(8) [6 marks]

```
function find_x(bmt, x) {

  return is_empty_tree(bmt)
         ? false
         : x === entry(bmt)
         ? true
         :
}
```

(9) [5 marks]

```
function flatten(bmt) {
  return

}
```

(10) [8 marks]



(11) [8 marks]

Environment of Program D1:

(12) [8 marks]

Environment of Program D2:

(13) [6 marks]

```
function utm(n) {
    const M = [];
    for (let r = 0; r < n; r = r + 1) {
        M[r] = [];
        for (let c = 0; c < n; c = c + 1) {



        }
    }
    return M;
}
```

(14) [3 marks] ○ **A** ○ **B** ○ **C** ○ **D** ○ **E** ○ **F**

(15) [3 marks] ○ **A** ○ **B** ○ **C** ○ **D** ○ **E** ○ **F**

(16) [3 marks] ○ **A** ○ **B** ○ **C** ○ **D** ○ **E** ○ **F**

(17) [3 marks] ○ **A** ○ **B** ○ **C** ○ **D** ○ **E** ○ **F**

(18) [5 marks]

```
function stream(...xs) {
    const n = array_length(xs);

}
```

(19) [3 marks]

```
function lazier_stream_element(s, n) {

}
```

(20) [5 marks]

```
function valid_next(row, col, maze) {

    const n_rows = array_length(maze);    // number of rows
    const n_cols = array_length(maze[0]); // number of columns

    const next = [[row, col+1], [row+1, col], [row+1, col+1]];
    let n_valids = 0;
    const ans = [];

    for (let i = 0; i < array_length(next); i = i + 1) {




    }

    return ans;
}
```

(21) [6 marks]

```
function is_solvable(row, col, maze) {

    const n_rows = array_length(maze);    // number of rows
    const n_cols = array_length(maze[0]); // number of cols
    let ans = false;

    if (                                                  ) {

        // base case – have reached end

        ans = true;

    } else {

        // have not reached end yet

        const next = valid_next(row, col, maze);

        for (let i = 0; i < array_length(next); i = i + 1) {




        }
    }

    return ans;
}
```

——— END OF ANSWER SHEET ———